

YANCHI, China

WMO: 537230

Lat: 37.7925N

Lon: 107.3899E

Elev: 1356

StdP: 86.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-17.1	-28.6	0.3	-13.0	-26.4	0.4	-11.6	9.3	-0.6	8.3	-1.3	0.7	270	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.4	16.6	31.1	16.4	29.7	16.2	20.7	26.5	19.9	25.8	19.2	25.1	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	16.5	23.1	18.2	15.5	22.7	17.2	14.5	22.2	67.1	26.5	63.8	25.8	60.9	25.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.2	6.2	DB	-22.8	34.9	2.8	1.4	-24.8	35.9	-26.4	36.7	-28.0	37.5	-30.1	38.5
			WB	-23.2	21.8	2.7	1.1	-25.2	22.5	-26.7	23.2	-28.2	23.8	-30.2	24.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-7.0	-3.0	4.0	11.2	16.6	21.1	23.2	21.5	16.2	9.6	2.0	-5.0
	DBStd	11.12	4.40	4.90	5.03	4.65	3.65	2.81	2.72	2.96	3.46	3.91	4.41	4.28
	HDD10.0	1889	528	363	194	40	2	0	0	0	2	54	241	466
	HDD18.3	3706	786	596	445	219	78	8	1	7	81	271	490	724
	CDD10.0	1618	0	0	8	75	206	334	410	356	187	42	1	0
	CDD18.3	393	0	0	0	4	23	92	153	104	17	0	0	0
	CDH23.3	3951	0	0	2	70	353	1027	1455	878	159	7	0	0
	CDH26.7	1329	0	0	0	16	77	353	577	273	34	0	0	0
Wind	WSAvg	2.5	2.2	2.4	2.8	3.0	2.9	2.7	2.5	2.3	2.3	2.2	2.5	2.4
Precipitation	PrecAvg	278	2	3	9	18	27	30	49	74	42	20	8	1
	PrecMax	488	11	10	56	83	74	81	110	234	95	47	29	5
	PrecMin	145	0	0	0	0	0	6	14	8	6	2	0	0
	PrecStd	80	2	3	11	18	19	19	23	50	24	12	9	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	14.3	22.5	28.8	30.7	33.4	35.1	32.8	30.8	24.2	17.7	9.9
		MCWB	0.9	4.1	8.6	12.1	13.0	14.9	17.7	17.7	14.4	11.5	6.9	1.3
	2%	DB	5.9	10.4	18.4	25.3	28.7	31.7	33.0	31.1	26.9	21.6	14.4	6.7
		MCWB	-1.1	2.0	6.6	10.4	12.6	14.9	17.7	17.7	14.9	11.0	5.3	-0.3
	5%	DB	3.2	7.7	15.6	22.6	27.1	30.3	31.5	29.6	24.9	19.3	12.1	4.3
		MCWB	-2.5	0.6	5.2	9.2	12.2	14.7	17.3	17.7	14.2	9.9	4.2	-1.4
	10%	DB	0.8	5.7	13.1	20.4	25.3	28.8	30.0	28.1	23.1	17.2	9.8	2.2
		MCWB	-3.9	-0.4	4.0	8.3	11.4	14.4	17.2	17.2	14.1	9.0	3.3	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	4.8	9.4	13.7	15.8	19.4	22.1	21.8	19.7	14.1	8.2	2.1
		MCDB	8.4	13.1	19.9	20.3	24.5	25.2	27.4	28.0	22.6	20.6	13.8	7.8
	2%	WB	-0.9	2.7	7.3	12.0	14.6	18.2	20.9	20.7	17.6	12.5	6.6	0.2
		MCDB	5.3	9.2	16.6	21.8	23.6	25.0	27.2	26.1	22.5	18.2	12.0	6.2
	5%	WB	-2.2	1.3	5.8	10.7	13.7	17.2	20.1	19.9	16.3	11.3	5.2	-1.1
		MCDB	3.0	6.8	14.4	19.4	22.5	24.1	26.5	25.4	21.2	17.0	10.5	3.7
	10%	WB	-3.7	-0.1	4.4	9.6	12.8	16.3	19.4	19.2	15.3	10.2	3.9	-2.4
		MCDB	0.6	5.2	11.9	17.8	21.6	23.4	25.7	24.9	20.3	15.4	9.2	2.1
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.5	13.3	13.5	13.0	12.4	11.0	10.6	10.8	12.3	12.3	12.2
		MCDBR	16.1	16.8	17.2	16.9	16.4	15.4	13.7	12.8	14.0	14.9	15.4	14.8
		MCWBR	11.0	10.6	9.2	7.4	6.2	4.5	3.6	3.6	4.9	6.6	8.6	9.8
	5% WB	MCDBR	15.5	15.3	15.3	13.4	12.6	10.3	9.8	9.5	9.6	12.2	13.4	13.8
		MCWBR	10.8	10.0	8.6	6.4	5.5	3.9	3.4	3.4	4.2	6.0	8.1	9.4
Clear-Sky Solar Irradiance	taub	0.351	0.416	0.465	0.500	0.513	0.526	0.505	0.517	0.489	0.448	0.375	0.337	
	taud	2.033	1.883	1.807	1.765	1.803	1.837	1.974	1.916	1.924	1.940	2.037	2.105	
	Ebn at Noon	793	781	784	785	783	772	785	763	756	739	758	783	
	Edh at Noon	125	166	198	219	214	207	180	186	174	155	123	108	
All-Sky Solar Radiation	RadAvg	2.76	3.58	4.84	5.83	6.46	6.44	6.05	5.43	4.37	3.85	3.03	2.55	
	RadStd	0.26	0.29	0.31	0.41	0.33	0.41	0.31	0.34	0.37	0.33	0.31	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	-0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page